

Простые сети в GNS3. Анализ трафика

Лабораторная работа №5

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29 октября 2025

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Цели и задачи работы

Построение простейших моделей сети на базе коммутатора и маршрутизаторов **FRR** и **VyOS** в **GNS3**, анализ сетевого трафика с помощью **Wireshark**.

1. Смоделировать сеть с коммутатором и узлами **VPCS**.
2. Настроить IP-адресацию и проверить связность.
3. Выполнить анализ трафика **ARP, ICMP, UDP** и **TCP**.
4. Настроить маршрутизаторы **FRR** и **VyOS**.
5. Оценить работу сетевых протоколов на канальном и сетевом уровнях.

Теоретическая часть

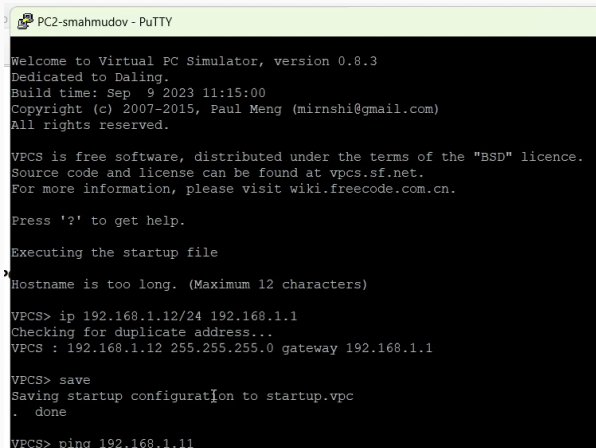
- **GNS3** — эмулятор сетевой инфраструктуры.
- **FRRouting (FRR)** — реализация протоколов маршрутизации: OSPF, BGP, RIP и др.
- **VyOS** — дистрибутив Linux, выполняющий функции маршрутизатора и брандмауэра.
- **VPCS** — эмулятор конечных устройств.
- **Wireshark** — анализатор сетевых пакетов.

- **ARP** — сопоставляет IP-адреса с MAC-адресами в локальной сети.
- **ICMP** — используется для проверки доступности узлов (ping).
- **UDP** — протокол без установления соединения.
- **TCP** — обеспечивает надёжную доставку данных (SYN, ACK, FIN).

Выполнение лабораторной работы

Настройка IP и проверка связи

- PC1: ip 192.168.1.11/24 192.168.1.1
- PC2: ip 192.168.1.12/24 192.168.1.1
- Проверка командой `ping 192.168.1.11` — ответ получен.



```
PC2-smahmudov - PuTTY

Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file
>
Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> ping 192.168.1.11
```

- В Wireshark зафиксированы Gratuitous ARP-пакеты.
- Подтверждена уникальность IP-адресов устройств.

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	::	ff02::2	ICMPv6	62	Router Solicitation
2	0.001940	::	ff02::2	ICMPv6	62	Router Solicitation
3	0.050549	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
4	0.052240	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
5	1.051134	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
6	1.053410	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
7	2.051173	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
8	2.054242	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)

> Frame 3: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface 0

> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Address Resolution Protocol (request/gratuitous ARP)

Hardware type: Ethernet (1)
 Protocol type: IPv4 (0x0800)
 Hardware size: 6
 Protocol size: 4
 Opcode: request (1)
 [Is gratuitous: True]
 Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
 Sender IP address: 192.168.1.11
 Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
 Target IP address: 192.168.1.11

0000 ff ff ff ff ff ff 00 50 79 66 68 00 08 06 00 01
 0010 08 00 06 04 00 01 00 50 79 66 68 00 c0 a8 01 0b
 0020 ff ff ff ff ff ff c0 a8 01 0b 00 00 00 00 00 00
 0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

- При выполнении ping видны Echo Request/Reply.
- Связь функционирует корректно.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	::	ff02::2	ICMPv6	62	Router Solicitation
2	0.001940	::	ff02::2	ICMPv6	62	Router Solicitation
3	0.050549	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
4	0.052240	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
5	1.051134	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
6	1.053410	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
7	2.051173	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
8	2.054242	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
9	100.440648	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.11? Tell 192.168.1.12
10	100.441782	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00
→ 11	100.443749	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x1d66, seq=1/256, ttl=64 (
← 12	100.444455	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x1d66, seq=1/256, ttl=64 (


```

> Frame 11: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
> Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11
v Internet Control Message Protocol
  Type: 8 (Echo (ping) request)
  Code: 0
  Checksum: 0x02a5 [correct]
  [Checksum Status: Good]
  Identifier (BE): 7526 (0x1d66)
  Identifier (LE): 26141 (0x661d)
  Sequence Number (BE): 1 (0x0001)
  Sequence Number (LE): 256 (0x0100)
  [Response frame: 12]
> Data (56 bytes)
0000 00 50 79 66 68 00 00 50 79 6
0010 00 54 66 1d 00 00 40 01 91 2
0020 01 0b 08 00 02 a5 1d 66 00 0
0030 0e 0f 10 11 12 13 14 15 16 1
0040 1e 1f 20 21 22 23 24 25 26 2
0050 2e 2f 30 31 32 33 34 35 36 3
0060 3e 3f

```

- Захвачены UDP-пакеты (порт назначения 7).
- Зафиксировано трёхстадийное установление TCP-сессии: SYN, SYN-ACK, ACK.
- Завершение — через FIN-ACK.

No.	Time	Source	Destination	Protocol	Length	Info
11	100.443749	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x1d66, seq=1/256, ttl=64
12	100.444455	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x1d66, seq=1/256, ttl=64
13	161.159916	192.168.1.12	192.168.1.11	ECHO	98	Request
14	161.160463	192.168.1.11	192.168.1.12	ECHO	98	Response
15	204.919699	192.168.1.12	192.168.1.11	TCP	74	21750 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSV
16	204.920106	192.168.1.11	192.168.1.12	TCP	54	7 → 21750 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
17	204.921575	192.168.1.12	192.168.1.11	TCP	66	21750 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSV
18	204.923031	192.168.1.12	192.168.1.11	ECHO	122	Request
19	204.923421	192.168.1.11	192.168.1.12	TCP	54	7 → 21750 [ACK] Seq=1 Ack=57 Win=2920 Len=0
20	204.926096	192.168.1.12	192.168.1.11	TCP	66	21750 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920 L
21	204.926437	192.168.1.11	192.168.1.12	TCP	54	7 → 21750 [ACK] Seq=1 Ack=58 Win=2920 Len=0
22	204.926479	192.168.1.11	192.168.1.12	TCP	54	7 → 21750 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
23	204.929649	192.168.1.12	192.168.1.11	TCP	66	21750 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSV

```

> Frame 15: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:
> Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11
> Transmission Control Protocol, Src Port: 21750, Dst Port: 7, Seq: 0, Len: 0
  Source Port: 21750
  Destination Port: 7
  [Stream index: 0]
  [Stream Packet Number: 1]
  > [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 340739166
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 0
  Acknowledgment number (raw): 0
  0000 00 50 79 66 68 00 00 50 79 6
  0010 00 3c 66 66 00 00 40 06 90 c
  0020 01 0b 54 f6 00 07 14 4f 44 5
  0030 0b 68 3f 0e 00 00 02 04 05 b
  0040 66 66 00 00 00 00 01 03 03 0

```

- Топология: FRR, коммутатор и VPCS.
- Маршрутизатор: 192.168.1.1/24.
- Узел PC1: 192.168.1.10/24.

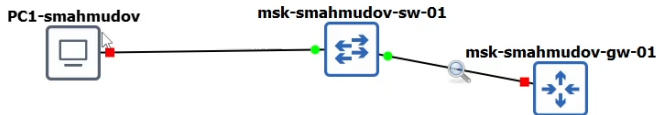


Рис. 5: Топология FRR

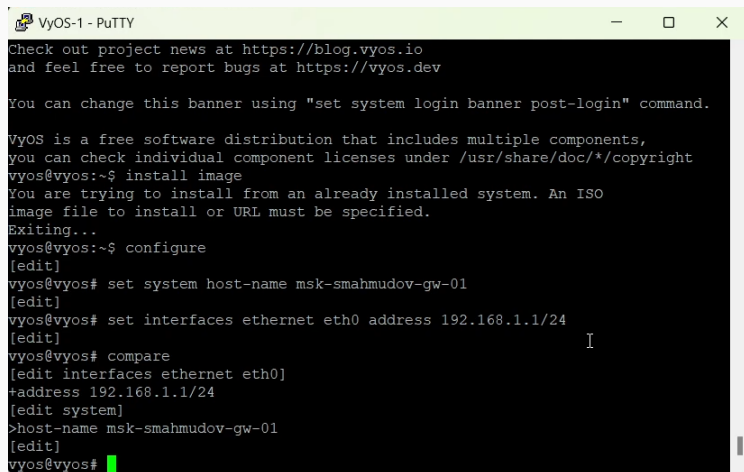
Проверка связи и анализ

- Ping между PC1 и маршрутизатором успешен.
- В Wireshark зафиксированы ARP и ICMP Echo пакеты.

No.	Time	Source	Destination	Protocol	Length	Info
13	197.900859	0c:0f:80:fa:00:00	Private_66:68:00	ARP	60	192.168.1.1 is at 0c:0f:80:fa:00:00
14	197.902305	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x5668, seq=1/256, ttl=64
15	197.903154	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x5668, seq=1/256, ttl=64
16	198.903864	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x5768, seq=2/512, ttl=64
17	198.904689	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x5768, seq=2/512, ttl=64
18	199.905114	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x5868, seq=3/768, ttl=64
19	199.905619	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x5868, seq=3/768, ttl=64
20	200.906523	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x5968, seq=4/1024, ttl=64
21	200.907298	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x5968, seq=4/1024, ttl=64
22	201.908583	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x5a68, seq=5/1280, ttl=64
23	201.909601	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x5a68, seq=5/1280, ttl=64
24	202.910606	0c:0f:80:fa:00:00	Private_66:68:00	ARP	60	Who has 192.168.1.10? Tell 192.168.1.1
25	202.910829	Private_66:68:00	0c:0f:80:fa:00:00	ARP	60	192.168.1.10 is at 00:50:79:66:68:00

> Frame 1: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface -, id 0	0000	33 33 00 00 02 00 50 79 66 68 00 00 00 00 00 00
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: IPv6mcast_02 (33:33:00:00:00:02)	0010	00 00 00 08 3a ff 00 00 00 00 00 00 00 00 00 00
> Internet Protocol Version 6, Src: ::, Dst: ff02::2	0020	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
> Internet Control Message Protocol v6	0030	00 00 00 00 00 02 85 00 7b 6b 00 00 00 00 00 00

- Установлено имя узла `msk-smahmudov-gw-01`.
- Интерфейс `eth0`: `192.168.1.1/24`.

A screenshot of a terminal window titled "VyOS-1 - PuTTY". The terminal displays the VyOS login banner and a series of configuration commands. The user enters 'install image', which prompts for an ISO file or URL. Then, the user enters 'configure', which enters the configuration editor. Inside the editor, the user sets the host name to 'msk-smahmudov-gw-01' and the IP address for interface 'eth0' to '192.168.1.1/24'. The user then enters 'compare' to show the configuration changes, followed by '>' to save them. The terminal ends with the prompt 'vyos@vyos#'.

```
VyOS-1 - PuTTY
Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ install image
You are trying to install from an already installed system. An ISO
image file to install or URL must be specified.
Exiting...
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-smahmudov-gw-01
[edit]
vyos@vyos# set interfaces ethernet eth0 address 192.168.1.1/24
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
+address 192.168.1.1/24
[edit system]
>host-name msk-smahmudov-gw-01
[edit]
vyos@vyos#
```

Проверка соединения

- PC1: 192.168.1.10/24 192.168.1.1
- Проверка ping 192.168.1.1 — ответы получены.
- В Wireshark зафиксированы ARP и ICMP пакеты.

No.	Time	Source	Destination	Protocol	Length	Info
33	371.973944	::	ff02::16	ICMPv6	130	Multicast Listener Report Message v2
34	372.713134	::	ff02::16	ICMPv6	130	Multicast Listener Report Message v2
35	372.777336	::	ff02::1:ffd8:0	ICMPv6	86	Neighbor Solicitation for fe80::e50:b0ff:fed8:0
36	373.803793	fe80::e50:b0ff:fed8...	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
37	373.806771	fe80::e50:b0ff:fed8...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
38	374.505278	fe80::e50:b0ff:fed8...	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
39	374.760867	fe80::e50:b0ff:fed8...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
40	419.689830	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.10
41	419.692559	0c:50:b0:d8:00:00	Private_66:68:00	ARP	60	192.168.1.1 is at 0c:50:b0:d8:00:00
42	419.694041	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xff6a, seq=1/256, ttl=64
43	419.696371	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xff6a, seq=1/256, ttl=64
44	420.697362	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x006b, seq=2/512, ttl=64
45	420.698817	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x006b, seq=2/512, ttl=64
46	421.700786	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x016b, seq=3/768, ttl=64
47	421.702583	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x016b, seq=3/768, ttl=64
48	422.704863	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x026b, seq=4/1024, ttl=64
49	422.706342	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x026b, seq=4/1024, ttl=64
50	423.707634	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x036b, seq=5/1280, ttl=64
51	423.709866	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x036b, seq=5/1280, ttl=64
52	424.806296	0c:50:b0:d8:00:00	Private_66:68:00	ARP	60	Who has 192.168.1.10? Tell 192.168.1.1
53	424.807660	0c:50:b0:d8:00:00	0c:50:b0:d8:00:00	ARP	60	192.168.1.10 is at 0c:50:b0:d8:00:00

> Frame 1: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface -, id 0

> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: IPv6mcast_02 (33:33:00:00:00:02)

> Internet Protocol Version 6, Src: ::, Dst: ff02::2

> Internet Control Message Protocol v6

0000

0010

0020

0030

33 33 00 00 00 02 00 50 79 66 68 00 00 00 00 00

0000

0010

0020

0030

00 00 00 00 00 00 ff 02 00 00 00 00 00 00 02 85 00 7b b1

Вывод

- Смоделированы сети на базе **FRR** и **VyOS**.
- Настроена IP-адресация и проверена связность.
- Проанализированы пакеты **ARP, ICMP, UDP, TCP** в **Wireshark**.
- Подтверждена корректная работа сетевых протоколов и взаимодействие устройств в **GNS3**.